

Past, Current and Future Trends in System Analysis and Design

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Abstract— Past, present and future trends in systems analysis and design were researched that cover the background of the methodologies and technologies used in system development. Their definition and key factors of why they were popular in their respective times were also researched.

Keywords— Waterfall, Object-Oriented Analysis and Design (OOAD), Agile, DevOps, Artificial Intelligence (AI), Internet of Things (IoT).

I. INTRODUCTION

System analysis and design is a process used by many businesses to examine specific company conditions and propose strategies for improving them using more efficient approaches. Companies may utilise this method to upscale their organisation or to achieve corporate growth and profitability goals. System analysis and design often emphasise how systems behave, how they interact with other subsystems, and their ability to achieve a certain purpose. This frequently entails examining a system's performance and the quality of output. Over the years, more and more technologies and methodologies have been developed to further improve the efficiency of systems analysis and design. With each new technology, it improves the previous approach of system analysis and design while also magnifying its capabilities. This paper will discuss the trends in systems analysis and design throughout time.

II. PAST TRENDS

A. Waterfall Model

The waterfall methodology was first introduced by Winston W. Royce in 1970[1]. It was noted that it was in one of his papers titled "Managing the Development of Large Software Systems". Royce described a software development paradigm using a procedural model with six stages: requirements analysis, design, implementation, verification, integration, and maintenance. By using the waterfall methodology, the goal is set rather early on in to facilitate the work in a goal- and plan-oriented manner [2]. It is organised comprehensively, with tasks, roles, and due dates, from the beginning to the end. The goal is to carry out the original plan as closely as possible. This offers predictable resources, documented planning, and stability and structure. However, the flaw that comes with the model is that if a task remains undone, the stage cannot proceed

[3]. This will cause additional costs than what was planned due to more discussions with the client to clarify their needs. The flaw translates to decreasing trends throughout time as other methodologies are seen to be more viable.

B. Object-Oriented Analysis and Design

Object-oriented analysis and design (OOAD) were first introduced in the 1960s but gained popularity in the late 1990s because it increased developer productivity and software quality [4]. It is a powerful model that works by analysing and collecting user requirements, as well as developing applications using visual aids to improve software [5]. Using OOAD effectively improves software maintenance, reusability, and modifiability. However, the drawback of using this model is that has limited support for functional analysis, mostly because UML representations are limited in large-scale projects [6].

III. PRESENT TRENDS

A. Agile Methodology

Agile methodologies are one of the most used methods around the world. Over 20 different agile methodologies exist some of which are Scrum, extreme programming and Kanban [7]. Agile is sometimes described as iterative and incremental, aiming to avoid traditional methodologies that emphasise early design and specification freeze, a fixed project scope, and little customer engagement. Essentially, it is the ability to adapt to changes during product development. This is in contrast with the traditional approach where it is focused more towards the planning instead of the execution. As mentioned, when developing a system with the waterfall methodology, if a phase is not done, then the development cannot proceed. Hence, the agile method became popular because it embraces changes as changes are inevitable in system development. One of the downsides of this methodology is it heavily revolves around good communication and collaboration between team members and clients [8], [9]. It also often ignores software delivery and deployment [10].

B. DevOps

DevOps was first introduced in 2009, as an integration of development and operations tasks in a team [10]. DevOps'

purpose is to encourage collaboration between development and operations, in part by automating tasks for assembling and deploying code and testing, to reduce the time taken in software development, continuation of software delivery, and reduce software instability thus improving satisfaction of customers [10]. Through the combination of development and operations, it provides flexibility in rapidly changing situations to fulfil the client's requirements. DevOps also shares the same drawback of agile methodology. What sets them apart is DevOps overcomes a weakness of the agile methodology which is the neglect of software delivery and deployment [10].

IV. FUTURE TRENDS

A. Artificial Intelligence (AI) Development

The meaning of Artificial intelligence refers to the capacity of a computer or a robot under computer control to carry out operations that are typically associated with human intellectual processes, including reasoning [11]. The term is frequently associated with technologies of developing systems that mimic human behaviour mainly, intellectual processes such as the ability or learn from experience.

Systems analysis has been transformed due to the power of AI. AI algorithms and machine learning models can now analyse huge volumes of data at a faster pace compared to a human analyst [11]. By leveraging advanced analytics techniques, AI can identify patterns, trends, and anomalies that humans might overlook, leading to more accurate and comprehensive analysis.

1) *Enhanced efficiency*: With the ability of AI that process data with an incredible speed, system analysis now can be completed within a short amount of time. This will allow many users or organizations to identify and able to address issues instantly, improving overall efficiency and reducing the rate of management [11].

2) *Improved accuracy*: AI can eliminate the risk of human error, and can make an analysis more accurate and reliable. By filtering many unnecessary elements from the process, users or organizations can confidently use a conclusion from the data [11].

3) *Data-driven insights*: AI tools can analyse vast amounts of data to provide valuable insights into user behaviour, market trends and design performance. This data-driven approach helps designers make informed decisions [11].

B. Internet of Things (IoT)

One new yet important topic in technology, society, and economy is the Internet of Things (IoT). When internet connectivity and robust data analytic skills are joined with consumer goods, durable goods, automobiles and trucks, industrial facilities and components, sensors, and other

commonplace objects, the result is a potential revolution in how humans live and work [12].

The power of the Internet of Things involves understanding how different components interact within a complex system. IoT has emerged as a powerful tool in systems analysis, providing a wealth of real-time data that was previously inaccessible. By connecting physical objects to the internet, IoT enables continuous monitoring and analysis of these objects, leading to valuable insights [12].

1) *Real-time data collection*: IoT devices continuously collect data from various sensors and devices embedded within a system. This real-time data allows analysts to gain a comprehensive understanding of the system at any given moment, enabling proactive decision-making [12].

2) *Increased visibility*: IoT provides visibility into previously opaque processes and systems. By monitoring physical objects and collecting relevant data, analysts can identify areas of improvement, optimize operational efficiencies, and enhance overall system performance [12].

3) *Predictive analytics*: With IoT, systems analysts can use powerful predictive analytics algorithms to forecast future trends and events. The ability to anticipate potential issues or predict demand allows organizations to be proactive and respond promptly, minimizing downtime and maximizing productivity [12].

IV. CONCLUSIONS

In conclusion, system analysis and design methodologies have evolved and advanced to deal with evolving levels of abstraction and the complexity of the business problems being modelled [12]. The field of system analysis and design has succeeded in providing methodologies for systems analysis, design, and development while not ignoring the practical and theoretical implications. The future of system analysis and design will involve the exploration of many research ideas and challenges that are highlighted in this paper.

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